PROJECT SYNOPSIS

Water Quality Classification and Contamination Risk Prediction

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1. Project Type

This is a Machine Learning based Data Science project that focuses on classification and predictive risk modeling for surface water bodies across India. The project uses real government water-monitoring data to classify monitoring stations as safe or contaminated and to predict the associated contamination risk level based on physicochemical and microbiological water quality parameters.

2. Industry Applications

The solution is designed for real-world deployment across sectors that depend on continuous surface water quality monitoring:

- Pollution Control Boards and environmental regulatory agencies (CPCB/SPCB-type monitoring)
- River, lake, canal, and pond water body management authorities
- Municipal and public health departments tracking water safety
- Marine and coastal monitoring (beach, sea, and creek water quality)
- Water Treatment Plant and STP raw-water intake assessment
- Research and policy-making bodies studying state-wise water pollution trends

3. Business Problem

India's surface water bodies — rivers, lakes, canals, ponds, drains, and coastal waters — are monitored across multiple states, but the resulting data is large, inconsistent, and reported as ranges (minimum-maximum) rather than single readings, with many missing values across parameters such as BOD, Nitrate-N, and coliform counts. Manually reviewing this station-wise data to judge whether a water body is safe or at contamination risk is slow and inconsistent across states and water body types. There is a need for an automated, data-driven system that can clean, standardize, and analyze this monitoring data to classify water quality and flag high contamination risk locations, enabling faster and more consistent decision-making by regulators.

4. Dataset Features

The dataset used for this project is a real Indian surface water quality monitoring dataset containing 194 monitoring station records spanning 2021 to 2023, collected across 17 states and covering 11 types of water bodies (River, Drain, Canal, Pond, Lake, Beach, Marine, Sea, Water Treatment Plant, STP, and Creek). Each record is identified by a station code and monitoring location, and includes the following parameters, most reported as Minimum and Maximum values:

- STN Code and Monitoring Location (station identifier and site name)
- Year of monitoring (2021-2023) and State Name
- Type of Water Body (River, Drain, Canal, Pond, Lake, Beach, Marine, Sea, STP, WTP, Creek)
- Temperature (°C) - Min/Max
- Dissolved Oxygen - Min/Max
- pH - Min/Max

- Conductivity (µmho/cm) - Min/Max
- BOD - Biochemical Oxygen Demand (mg/L) - Min/Max
- Nitrate-N (mg/L) - Min/Max
- Fecal Coliform (MPN/100ml) - Min/Max
- Total Coliform (MPN/100ml) - Min/Max
- Fecal Streptococci - Min/Max

A derived target label (e.g., Safe / Moderate Risk / High Risk) will be engineered from these parameters using established water quality standards (such as CPCB norms for BOD, coliform, and dissolved oxygen thresholds), since the raw dataset does not include a ready-made class label.

5. Machine Learning Workflow

The project follows a structured end-to-end machine learning pipeline tailored to this dataset's characteristics:

- Data collection and understanding of the 194-record, 23-column station-wise dataset
- Data cleaning: handling missing values in Conductivity, BOD, Nitrate-N, and coliform columns (13-102 missing entries), and resolving non-numeric entries such as 'BDL' (Below Detection Limit)
- Deriving representative values (e.g., mean of Min/Max) for each parameter per station
- Exploratory Data Analysis (EDA) across states, years, and water body types
- Feature engineering, encoding of categorical fields (State, Water Body Type), and feature scaling
- Construction of the target variable (contamination risk class) using water quality threshold rules
- Train-test split of the dataset
- Model building using classification algorithms (Logistic Regression, Decision Tree, Random Forest, XGBoost, etc.)
- Hyperparameter tuning and cross-validation
- Model evaluation and selection of the best-performing model
- Model saving and deployment readiness

6. Data Visualization Requirements

Visual analysis plays a key role in understanding patterns in this multi-state, multi-year dataset. The project includes:

- State-wise distribution of monitoring stations and average contamination levels
- Water body type comparison (River vs Drain vs Canal vs Lake vs Beach, etc.)
- Year-wise trend analysis (2021-2023) for key parameters like BOD and coliform counts
- Distribution and box plots for pH, Dissolved Oxygen, BOD, and Conductivity
- Correlation heatmap between water quality parameters
- Missing-value heatmap highlighting data gaps by parameter
- Class distribution chart (Safe vs Moderate Risk vs High Risk)
- Feature importance chart from the trained model
- Confusion matrix and ROC curve for model performance visualization

7. Evaluation Metrics

Model performance is assessed using standard classification evaluation metrics to ensure reliability and robustness, particularly given the moderate dataset size (194 records) and class imbalance across risk categories:

- Accuracy
- Precision, Recall, and F1-Score (weighted, to account for class imbalance)
- Confusion Matrix
- ROC-AUC Score
- Cross-validation score for generalization check

8. Final Deliverable

The final deliverable of the project includes a trained and validated machine learning model capable of classifying Indian surface water monitoring stations by contamination risk, a cleaned and documented version of the dataset, a complete project report with state-wise and water-body-wise insights, all visualizations, and source code. An interactive dashboard/application may also be provided to allow users to filter by state, water body type, or year, and view predicted contamination risk.

9. Advanced Extension

As a future enhancement, the project can be extended to include:

- Expanding the dataset with additional years and monitoring stations for improved generalization
- Real-time water quality monitoring using IoT sensors integrated with the trained model
- Geospatial mapping (state and river-basin level) to visualize contamination hotspots across India
- Deep learning or ensemble models for improved prediction accuracy on imbalanced classes
- Integration with cloud platforms for live data streaming and automated alerts to pollution control authorities